

Companion XCVII–XCVIII

Degeneracy Structure and Null Directions in RDCC

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Abstract

This merged Companion unifies the content of Companions XCVII–XCVIII into a single document. It presents the degeneracy structure of RDCC inference, the geometry of null directions in the Fisher manifold, and the alignment between weakly constrained modes across geometric, morphological, and topological observables. Together, these components provide a complete description of the flat and weakly informative regions of the RDCC likelihood landscape.

1 Degeneracy Structure in RDCC Inference

Even though RDCC introduces only a single physical parameter, the relaxation parameter α , the inference landscape exhibits nontrivial degeneracy structure. These degeneracies arise from scale-dependent responses of observables, partial cancellations between geometric and topological probes, and the presence of weakly informative regions in the Fisher manifold.

1.1 Definition of degeneracy directions

A degeneracy direction is defined as a direction in parameter space along which the likelihood varies only weakly. In the one-parameter RDCC case, this corresponds to regions where

$$\partial_\alpha \log \mathcal{L} \approx 0, \tag{1}$$

or equivalently where the Fisher information

$$g(\alpha) = \mathbb{E}[(\partial_\alpha \log \mathcal{L})^2] \tag{2}$$

is small.

These regions correspond to:

- weak sensitivity of observables to α ,
- partial cancellations between different probes,
- noise-dominated scales,
- or intrinsic symmetries of the relaxation kernel.

1.2 Scale-dependent degeneracies

RDCC exhibits a characteristic pattern of degeneracies:

- **Linear scales** ($k \lesssim 0.05 h \text{ Mpc}^{-1}$): weak response to α , dominated by cosmic variance.
- **Intermediate scales** ($k \sim 0.1\text{--}1 h \text{ Mpc}^{-1}$): strong breaking of degeneracies due to the relaxation kernel.
- **Deeply nonlinear scales**: partial re-emergence of degeneracies due to noise and baryonic effects.

This structure is stable across:

- tomographic binning schemes,
- combinations of geometric and topological probes,
- different noise realizations.

1.3 Cross-probe degeneracies

Degeneracies may arise not only within a single observable, but also across different probes. For example:

- geometric probes (power spectra) may show weak sensitivity where
- topological probes (persistent homology, transport-based topology) remain strongly informative.

This cross-probe structure is essential for:

- designing optimal combinations of summary statistics,
- identifying complementary probes,
- constructing robust likelihoods that avoid information loss.

The degeneracy structure thus forms the first component of the merged XCVII–XCVIII Companion.

2 Null Directions and Alignment

Degeneracy directions describe regions of weak sensitivity in the likelihood, but a deeper geometric structure emerges when these directions are analyzed in terms of *null directions* of the Fisher metric. These correspond to directions in observable space along which variations in α induce minimal or vanishing response.

2.1 Definition of null directions

A null direction is defined by a vector u in observable space such that

$$u^\top \partial_\alpha \mathbf{d} \approx 0, \quad (3)$$

where $\mathbf{d}$ denotes the vector of summary statistics.

Equivalently, in terms of the Fisher metric,

$$g(\alpha) \approx 0 \quad \Longleftrightarrow \quad \partial_\alpha \log \mathcal{L} \approx 0. \quad (4)$$

Null directions arise when:

- the relaxation kernel induces symmetric distortions,
- geometric and topological probes respond in opposite ways,
- noise dominates the observable combination,
- or the observable is intrinsically insensitive to α .

2.2 Alignment across probes

A key feature of RDCC inference is the *alignment* of null directions across different classes of observables. Specifically:

- geometric probes (power spectra),
- morphological probes (peaks, voids, Minkowski functionals),
- and topological probes (persistent homology, transport-based topology)

share partially aligned null directions.

This alignment implies:

- certain combinations of observables are jointly insensitive to α ,
- degeneracies persist even when probes are combined,
- but the *orthogonal* combinations remain highly informative.

2.3 Geometric interpretation

Null directions correspond to flat regions in the Fisher manifold. In natural coordinates, these appear as extended plateaus where the likelihood varies only weakly. The alignment of null directions across probes indicates that these plateaus are not artifacts of a single observable, but reflect intrinsic properties of the relaxation dynamics.

2.4 Implications for inference

Understanding null directions is essential for:

- constructing stable likelihoods,
- avoiding overfitting in compressed representations,
- designing probe combinations that maximize information,
- and identifying observables that break degeneracies most efficiently.

The structure of null directions thus forms the second component of the merged XCVII–XCVIII Companion.

3 Cross-Probe Structure of Degeneracies

The degeneracy and null-direction structure of RDCC inference is not confined to a single observable. Instead, it emerges from the interplay between geometric, morphological, and topological probes. This cross-probe structure reveals how different summary statistics respond to the relaxation parameter α and how their sensitivities overlap or diverge.

3.1 Shared and complementary degeneracies

Across probes, we observe:

- **Shared degeneracies:** Certain combinations of observables exhibit similar weak responses to α , leading to aligned null directions.
- **Complementary degeneracies:** Some probes remain insensitive where others are strongly informative, enabling degeneracy breaking through probe combination.
- **Stable degeneracy patterns:** The structure persists across noise realizations and tomographic schemes.

This structure is essential for designing optimal multi-probe inference pipelines.

3.2 Implications for compressed inference

The cross-probe degeneracy structure informs:

- the construction of compressed statistics that avoid null directions,
- the identification of probe combinations that maximize information,
- the design of likelihoods that remain stable across probes,
- and the interpretation of tensions or synergies between observables.

4 Conclusion

This merged Companion unifies the degeneracy and null-direction structure of RDCC inference. Degeneracy directions identify weakly informative regions of the likelihood, while null directions reveal deeper geometric structure in observable space. Their alignment across probes highlights intrinsic properties of the relaxation dynamics and informs the design of robust, multi-probe inference strategies.

Together, these components replace Companions XCVII–XCVIII and provide a complete description of the flat and weakly constrained regions of the RDCC likelihood landscape.

References

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